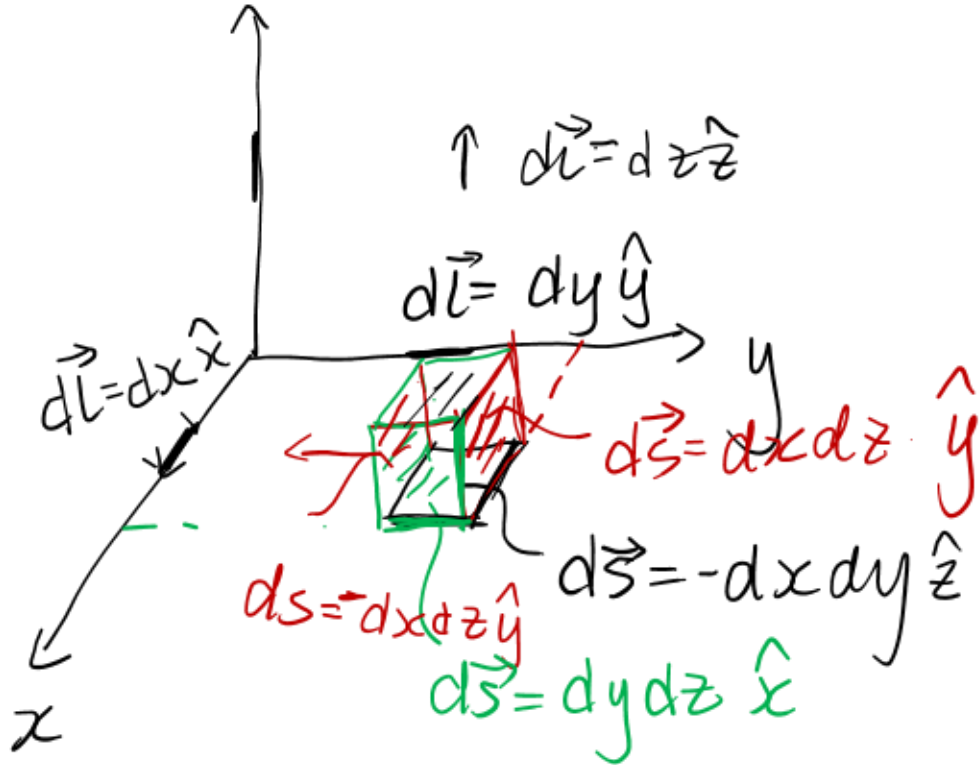




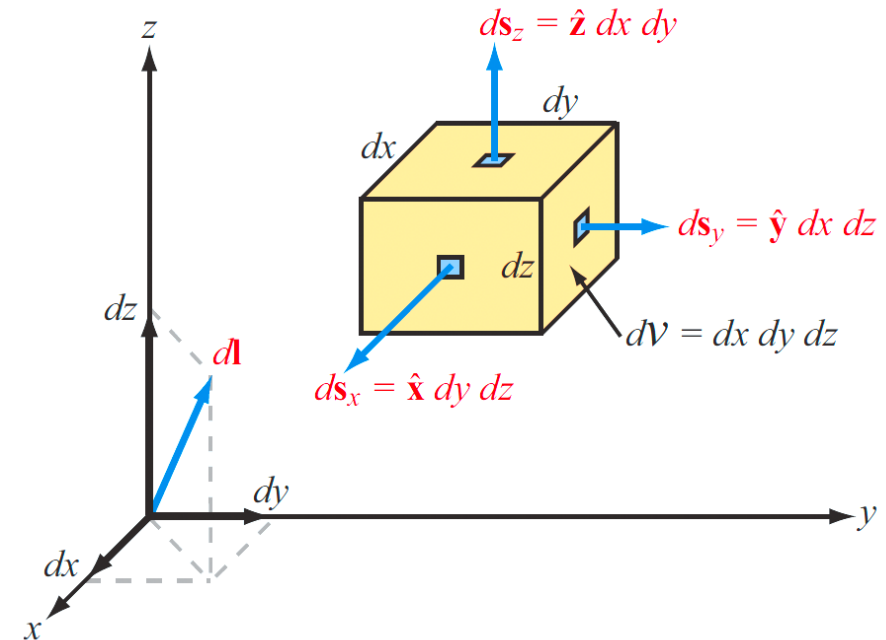
Lecture 4

- Differential length, surface, volume
- Charge, charge distribution
- Coulomb's law

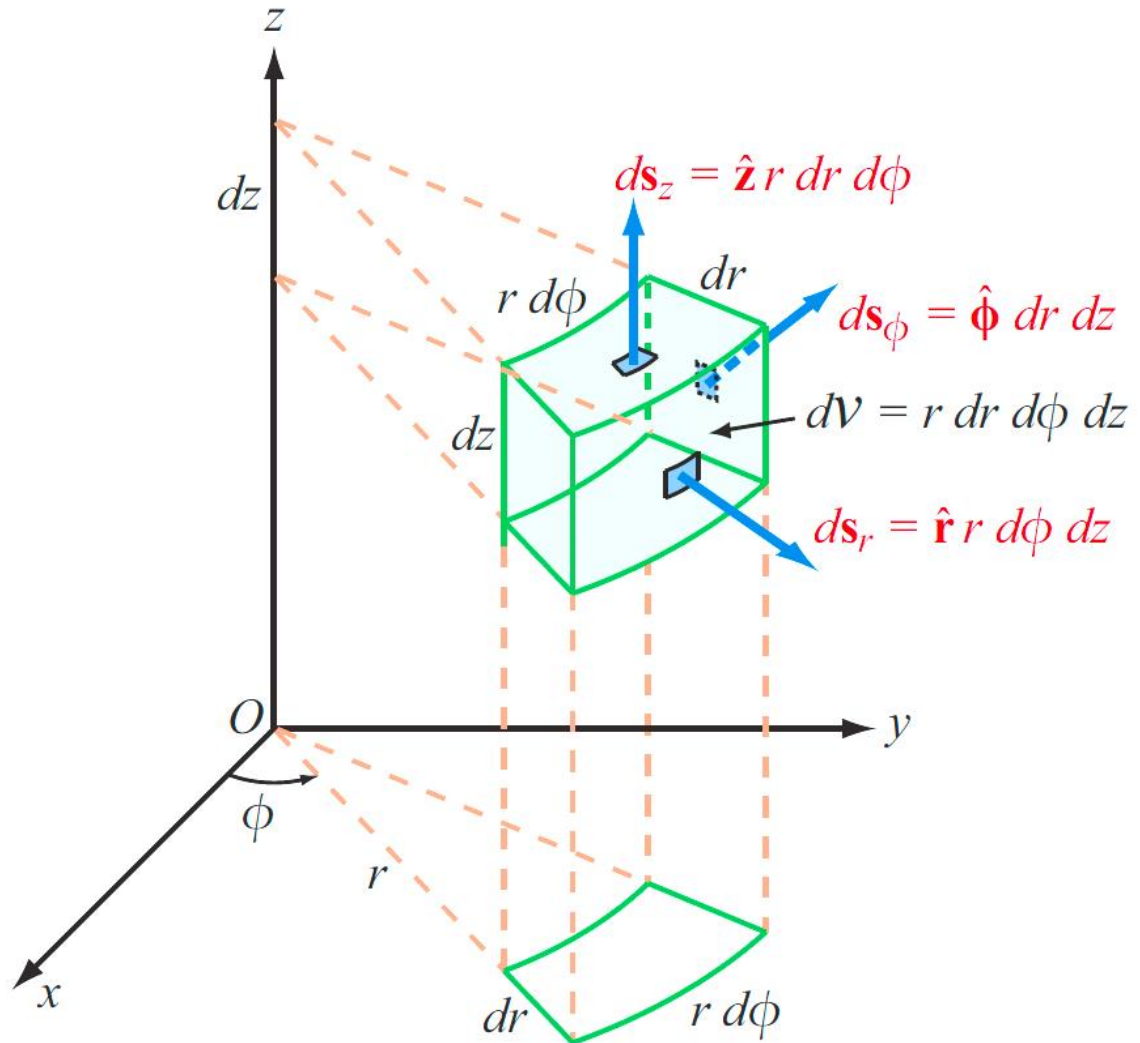
Differential length, surface, volume in Cartesian Coordinate System



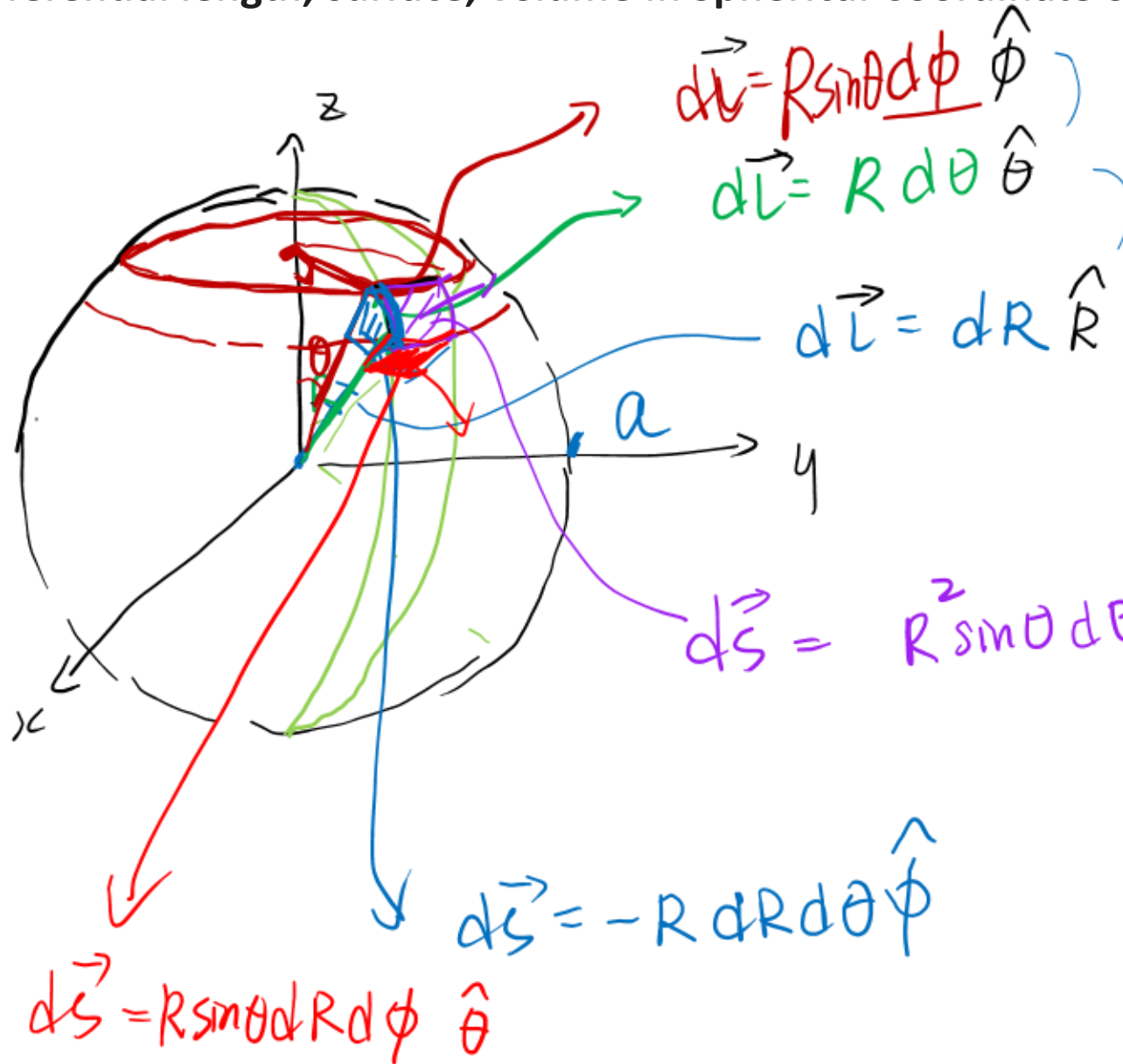
$$dV = dx dy dz$$



Differential length, surface, volume in Cylindrical Coordinate System



Differential length, surface, volume in Spherical Coordinate System



$$dV = R^2 \sin \theta dR d\theta d\phi$$

$$R \in [0, a]$$

$$\theta \in [0, \pi]$$

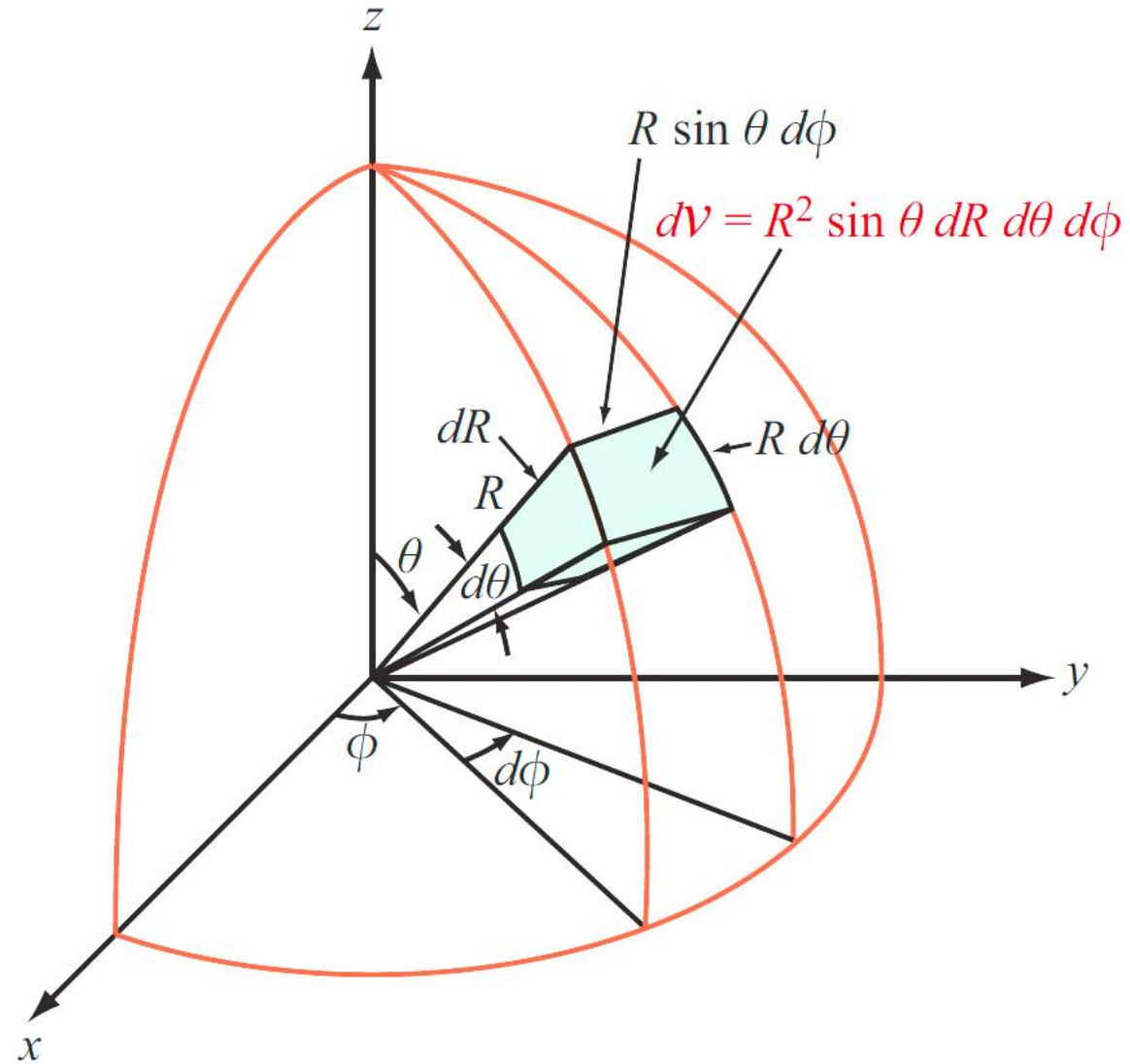
$$\phi \in [0, 2\pi)$$

$$\begin{aligned} V &= \int_{R=0}^{R=a} \int_{\theta=0}^{\theta=\pi} \int_{\phi=0}^{\phi=2\pi} \underline{R^2 \sin \theta dR d\theta d\phi} \\ &= \int_0^a R^2 dR \cdot \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \\ &= \frac{a^3}{3} \cdot 2 \cdot 2\pi = \frac{4\pi a^3}{3} \end{aligned}$$

Differential length, surface, volume in Spherical Coordinate System



A nice picture:



Electrostatics is the branch of physics that studies the forces, fields, and potentials produced by electric charges at rest (static charges).

Charge:
 → positive
 → negative

← ⊕

⊕ →

← ⊖

⊖ →

⊕ → ← ⊖



Fundamental properties of the electric charge



- **The law of conservation:** The electric charge can neither be created nor destroyed

In an isolated system, total charge Q is conserved.

- **Charge is quantized**

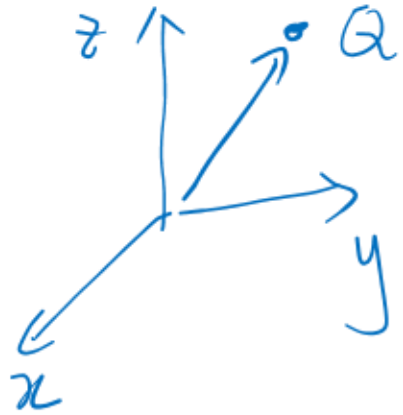
The charge of a single electron $q_e = -\frac{1.6 \times 10^{-19}}{e} \text{ C}$ Coulomb (C)

*of a proton
quark*

$$+e$$
$$-\frac{e}{3} ; +\frac{2e}{3}$$

For all the particle = integer multiple of $\frac{e}{3}$

Point charge and charge distribution



① line charge density $\rho_L = \frac{dQ}{dL} \left[\frac{C}{m} \right]$

② surface charge density $\rho_S = \frac{dQ}{dS} \left[\frac{C}{m^2} \right]$
 $\hookrightarrow Q = \int \rho_S dS$

③ volume charge density $\rho_V = \frac{dQ}{dV} \left[\frac{C}{m^3} \right]$

$\hookrightarrow Q = \int \rho_V dV$

$\rightarrow Q = \int \rho_V dV$



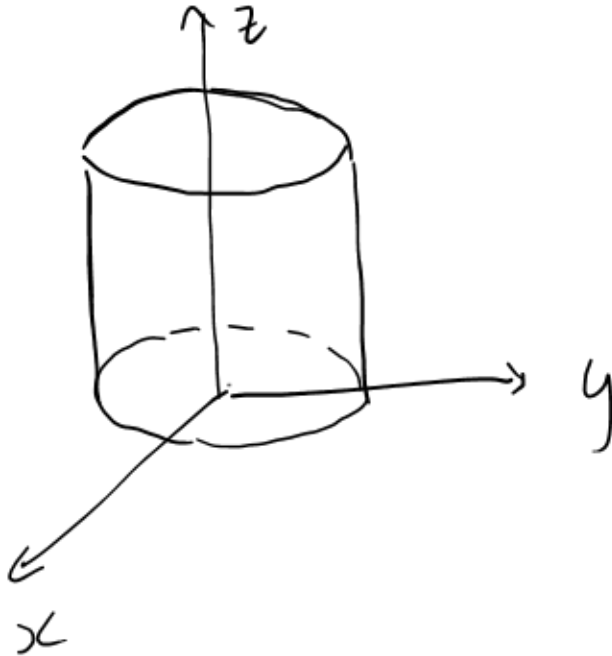
Point charge and charge distribution

4.8 An electron beam shaped like a circular cylinder of radius r_0 carries a charge density given by

$$\rho_v = \left(\frac{-\rho_0}{1+r^2} \right) \quad (\text{C/m}^3)$$

where ρ_0 is a positive constant and the beam's axis is coincident with the z -axis.

(a) Determine the total charge contained in length L of the beam.

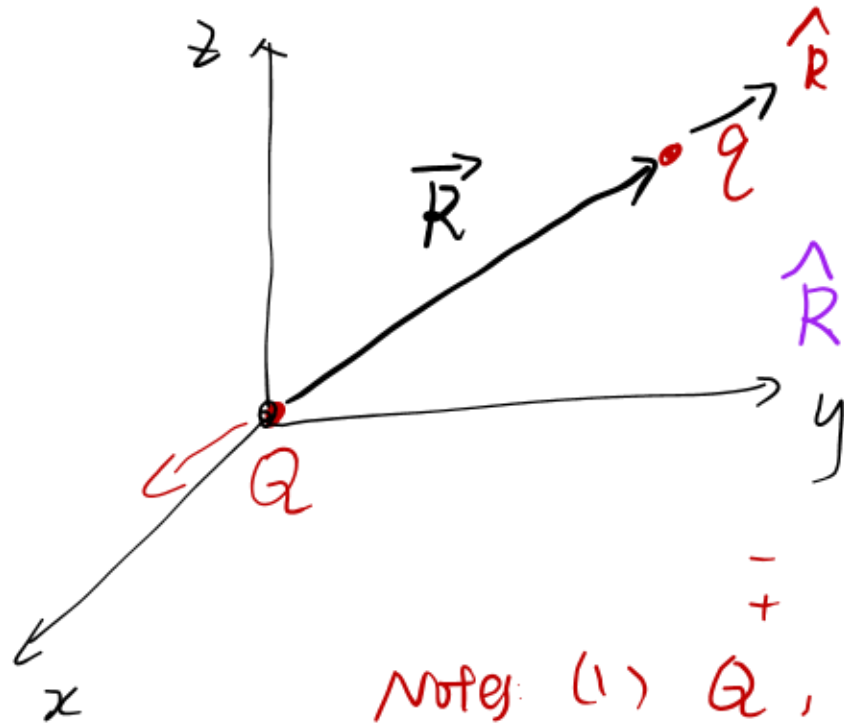


cylindrical coordinate system

let $u = 1+r^2$
 $du = 2r dr$
 $\frac{du}{2} = r dr$
 $\Rightarrow \int_1^{1+r_0^2} \frac{du}{2u} = \frac{1}{2} \ln(1+r_0^2)$

$$\begin{aligned} Q &= \int \rho_v dv = \int \rho_v r dr d\phi dz \\ r &\in [0, r_0] \\ \phi &\in [0, 2\pi) \\ z &\in [0, L] \\ &= -\rho_0 \int_0^{r_0} \frac{r}{1+r^2} dr \int_0^{2\pi} d\phi \int_0^L dz \\ &= -\rho_0 \left(\frac{1}{2} \ln(1+r_0^2) \right) \cdot 2\pi \cdot L \\ &= -\rho_0 \pi L \cdot \ln(1+r_0^2) \text{ C} \end{aligned}$$

Coulomb's law describes the amount of force between two electrically charged particles at rest.



$$\hat{R} = \frac{\vec{R}}{|\vec{R}|}$$

$$\vec{F}_{Q \rightarrow q}$$

$$= \frac{1}{4\pi\epsilon} \cdot \frac{Qq}{|\vec{R}|^2} \hat{R}$$

↑
permittivity

in free space $\epsilon = \epsilon_0 = 8.854 \times 10^{-12} \left[\frac{F}{m} \right]$
material $\epsilon = \epsilon_0 \epsilon_r$

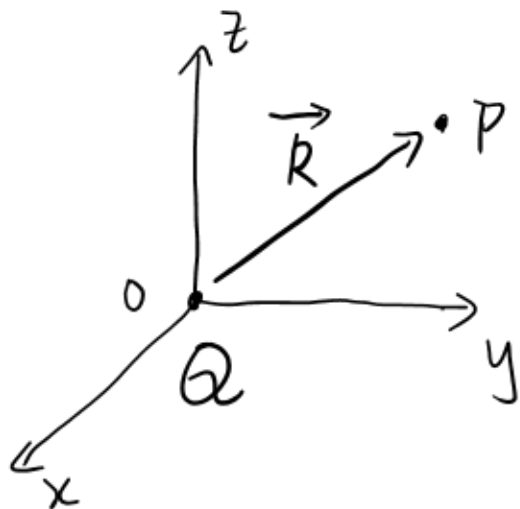
Notes: (1) $\begin{matrix} - & - & + \\ + & + & + \end{matrix}$ Q, q Qq force is repulsive

$\begin{matrix} + & - \\ + & - \end{matrix}$ Q, q Qq force is attractive

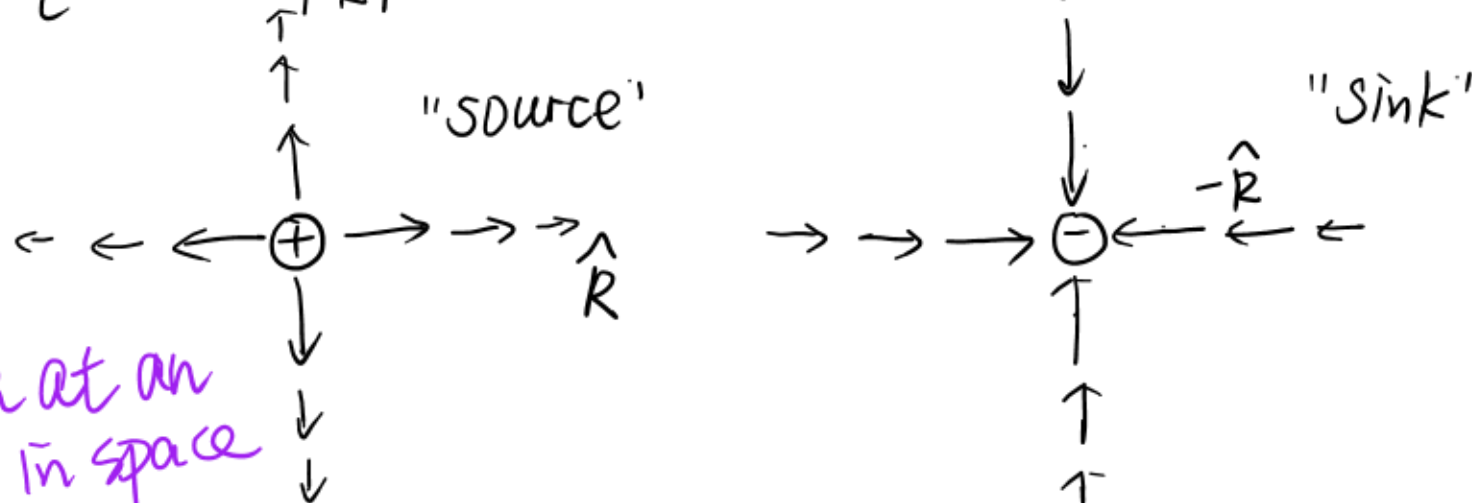
$$(2) \vec{F}_{q \rightarrow Q} = -\vec{F}_{Q \rightarrow q}$$

(3) Force decreases with distance $\frac{1}{|\vec{R}|^2}$

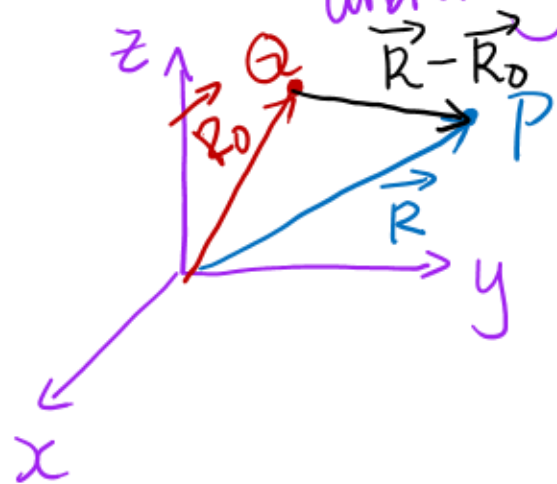
Coulomb's law for electric field



$$\vec{E} = \frac{\vec{F}}{q} = \frac{1}{4\pi\epsilon} \frac{Q}{|\vec{R}|^3} \vec{R} \quad \left[\frac{N}{C} \right]$$



Generalize to Q at an arbitrary point in space



(point of observation)

$\vec{E}$ produced by Q at P =

$$\vec{E} = \frac{1}{4\pi\epsilon} \frac{Q}{|\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0) \quad \left[\frac{N}{C} \right]$$

Source

point of observation

Source